Part3_Final

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Getting lists of parks

```
robotstxt::paths_allowed("https://stateparks.com/") #tests if we can scrape
```

```
[1] TRUE
```

```
session <- bow("https://stateparks.com/minnesota_parks_and_recreation_destinations.html", fo

print(session) # we use the polite package to scrape and add a crawl delay to
```

```
<polite session> https://stateparks.com/minnesota_parks_and_recreation_destinations.html
    User-agent: polite R package
    robots.txt: 4 rules are defined for 1 bots
   Crawl delay: 5 sec
  The path is scrapable for this user-agent
```

```
               # to follow website requirements

Sys.sleep(5)
parks_text <- scrape(session) |> #reads all of the text on the minnesota home
                                   # page
  html_nodes("#parklink a") |> #pulls a list of all parks and recreation
                               # destinations in minnesota
  html_text() #makes the list into a text list
```

Generating URLs

```
# formula -- generates urls we can scrape (only in mn)
#(adds the unique park name to the url for each list item)  --------------------
url_list <- vector("character")
for (i in 1:length(parks_text)) {
  form <- parks_text[i] |>
  str_to_lower() |>
  str_replace_all(c(" " = "_",
                    "'" = "",
                    "wildlife_mgmt." = "state_wildlife_management"))
  #for each park, the text is changed to lower case and put into snake case
  #(_ instead of spaces)
  #also takes care of differences between text displayed and urls
  url <- str_c("https://stateparks.com/", form, "_in_minnesota.html")
  #to create a list of the urls, this sets up the beginning and end of the url
  #(which is the same for every park)
  url_list <- c(url_list, url)
}
```

Scraping the first paragraph

```
# function using bow/scrape for finding paragraphs in parks
scrape_page <- function(url) {
  Sys.sleep(5) # this follows the website requests for a crawl delay
  session <- bow(url, force = TRUE) #setting polite session
  scrape(session) |>
    html_nodes("#overview :nth-child(5)") |>  #css selector for overview
    #css selector for the first paragraphs of the page
    html_text() # converts to text
}

# scraping the parks ------------------------------------------------------------
paragraph_list <- vector("character") # creates an empty character vector
for (i in 1:length(url_list)) { # iterates through the entire list of urls
  park_url <- url_list[i] # saves the current url
  paragraph <- scrape_page(park_url) # uses the scrape_page function above
  paragraph_list <- c(paragraph_list, paragraph)
  #appends the new paragraph to the list
}

head(paragraph_list)
```

```
[1] ""
[2] "Grand oaks and delicate prairie flowers grace the rugged, rolling landscape of this park
country skiers. The park offers visitors a swimming beach, backpack campsites and a visitor c
[3] ""
[4] "Treat yourself to a visit to Banning State Park, just minutes off I-
35 near Sandstone. In the spring, watch daring canoeists and kayakers shoot the turbulent rap
country skiing, beautiful trails for hiking and the study of natural history."
[5] ""
[6] "..."
```

```
#there's an extra space in between each paragraph that needs to be cleaned out
```

Cleaning paragraphs

```
park_par_tbl <- paragraph_list |>
  str_subset("[a-zA-Z]+") |> #removes strings that don't contain words
  str_subset("Email:|Picnickingyes", negate = TRUE)
#removes extra rows that were only recorded on some pages


# adding in missing rows -------------------------------------------------------
indexes <- c(9, 11, 20, 26, 37, 40, 107, 109, 110, 112, 113, 114, 115) #
#indexes of parks with missing paragraphs: cascade river, father hennepin,
#helmer myre, kadunce river, o l kipp, st croix wild river, boekholt grove,
#cooks, coon lake, little elk, malardi lake, talcot lake, teiken-dalve

for (i in 1:13) {
  park_par_tbl <- append(park_par_tbl, "", after = indexes[i])
} #adds in missing rows based on indexes of parks with missing paragraphs.
  #These will become NAs later.

head(park_par_tbl)
```

```
[1] "Grand oaks and delicate prairie flowers grace the rugged, rolling landscape of this park
country skiers. The park offers visitors a swimming beach, backpack campsites and a visitor c
[2] "Treat yourself to a visit to Banning State Park, just minutes off I-
35 near Sandstone. In the spring, watch daring canoeists and kayakers shoot the turbulent rap
country skiing, beautiful trails for hiking and the study of natural history."
```

```
[3] "Secluded in the Northwoods, this park contains pristine lakes; it is home to black bears
up with the Taconite State Trail and offer snowmobilers, skiers and hikers plenty to enjoy. 
bedroom guest house any season of the year."
[4] "Nestled in the blufflands of southeastern Minnesota, Beaver Creek Valley State Park is 
basswood and oak forest. Campers are lulled to sleep by the murmuring stream."
[5] "Big Stone Lake is 30 miles long and is located on the South Dakota-
Minnesota border. The lake is the source of the Minnesota River and attracts anglers who cat
[6] "Blue Mound State Park is on the tallest hill in southern Wisconsin, about 25 miles west
acre park is a popular place for swimming, hiking, camping, cross-country skiing and mountai
```

Scraping addresses

```r
# function for scraping addresses
scrape_address <- function(url) {
  Sys.sleep(5)
  session <- bow(url, force = TRUE)
  scrape(session) |>
    html_nodes(".parkinfo div:nth-child(1)") |>
    # css selector for the address box,
    # but it selects park name as well
    html_text()
}

# checking it works on individual urls
Sys.sleep(5)
session <- bow("https://stateparks.com/badoura_state_forest_in_minnesota.html",
               force = TRUE)
scrape(session) |>
  html_nodes(".parkinfo div:nth-child(1)") |> # css selector for address text
    html_text()
```

```
[1] "\n\n\nBADOURA STATE FOREST\n\n\t\n\t\n"
```

```r
# scraping addresses -- same form as function for scraping paragraphs
address_list <- vector("character")
for (i in 1:length(url_list)) {
  park_url2 <- url_list[i]
  address <- scrape_address(park_url2)
  address_list <- c(address_list, address)
}
```

```r
head(address_list)
```

```
[1] "\n\n\nAFTON STATE PARK6959 Peller Ave.South Hastings, Minnesota   55033\n\n\t\n\tPhone:
436-5391Toll Free: 888-646-6367 Email: \n"
[2] "\n\n\nBANNING STATE PARK61101 Banning Park RoadSandstone, Minnesota   55072\n\n\t\n\tPho
245-2668Toll Free: 888-646-6367 Email: \n"
[3] "\n\n\nBEAR HEAD LAKE STATE PARK9301 Bear Head State Park RoadEly, Minnesota   55731\n\n\
365-7229Toll Free: 888-646-6367 Reservations: 866-857-2757Email: \n"
[4] "\n\n\nBEAVER CREEK VALLEY STATE PARK15954 County 1Caledonia, Minnesota   55921\n\n\t\n\
724-2107Toll Free: 888-646-6367 Reservations: 866-857-2757Email: \n"
[5] "\n\n\nBIG STONE LAKE STATE PARK35889 Meadowbrook State Park RoadOrtonville, Minnesota
839-3663Toll Free: 888-646-6367 Reservations: 866-857-2757Email: \n"
[6] "\n\n\nBLUE MOUNDS STATE PARK1410 161st StreetLuverne, Minnesota   56156\n\n\t\n\tPhone:
283-1307Toll Free: 888-646-6367 Reservations: 866-857-2757Email: \n"
```

```r
# address list has a lot of extra characters to be cleaned
```

Cleaning addresses

```r
cleaned_addresses <- address_list |>
  # replace line breaks and tabs with spaces
  str_replace_all("[\r\n\t]", " ") |>
  #replace multiple spaces with a single space
  str_replace_all("  +", " ") |>
  #remove email addresses
  str_replace("Email:.*$", "") |>
  #remove phone numbers
  str_replace("Phone:.*$", "") |>
  #remove reservation info
  str_replace("Reservations:.*$", "") |>
  str_replace_all("^[^\\d]*", "") # removes the extra park name at the front

head(cleaned_addresses)
```

```
[1] "6959 Peller Ave.South Hastings, Minnesota   55033 "
[2] "61101 Banning Park RoadSandstone, Minnesota   55072 "
[3] "9301 Bear Head State Park RoadEly, Minnesota   55731 "
[4] "15954 County 1Caledonia, Minnesota   55921 "
[5] "35889 Meadowbrook State Park RoadOrtonville, Minnesota   56278 "
[6] "1410 161st StreetLuverne, Minnesota   56156 "
```

Making the final tidy tibble

```
#the initial tidy tibble
final_parks <- tibble(
  name = parks_text,
  url = url_list,
  address = cleaned_addresses,
  description = park_par_tbl
) |>
  mutate_all(na_if,"") #makes empty values in any column, any row, into NAs

# adding park type
final_parks <- final_parks |>
  mutate(type = case_when(
    #extended if/else logic
    str_detect(parks_text, "State Park") ~ "state park",
    str_detect(parks_text, "State Forest") ~ "state forest",
    str_detect(parks_text, "Mgmt. Area") ~ "wildlife management area",
    str_detect(parks_text, "National Park") ~ "national park",
    str_detect(parks_text, "National Forest") ~ "national forest",
    str_detect(parks_text, "Wildlife Refuge") ~ "wildlife refuge"
  ))

head(final_parks)
```

```
# A tibble: 6 x 5
  name                            url                    address description type
  <chr>                           <chr>                  <chr>   <chr>       <chr>
1 Afton State Park                https://stateparks.c~ "6959 ~ "Grand oak~ stat~
2 Banning State Park              https://stateparks.c~ "61101~ "Treat you~ stat~
3 Bear Head Lake State Park       https://stateparks.c~ "9301 ~ "Secluded ~ stat~
4 Beaver Creek Valley State Park https://stateparks.c~ "15954~ "Nestled i~ stat~
5 Big Stone Lake State Park       https://stateparks.c~ "35889~ "Big Stone~ stat~
6 Blue Mounds State Park          https://stateparks.c~ "1410 ~ "Blue Moun~ stat~
```

```
str(final_parks)
```

```
tibble [125 x 5] (S3: tbl_df/tbl/data.frame)
 $ name       : chr [1:125] "Afton State Park" "Banning State Park" "Bear Head Lake State Pa
 $ url        : chr [1:125] "https://stateparks.com/afton_state_park_in_minnesota.html" "http
```

```
 $ address    : chr [1:125] "6959 Peller Ave.South Hastings, Minnesota   55033 " "61101 Banni
 $ description: chr [1:125] "Grand oaks and delicate prairie flowers grace the rugged, rollin
35 near Sandstone. In the spring, watch dar"| __truncated__ "Secluded in the Northwoods, this
 $ type       : chr [1:125] "state park" "state park" "state park" "state park" ...
```

Making a .csv

```
write_csv(final_parks, "final_parks.csv") #writes the csv file
```